An Omitted Mode Is a Rare Rule: The Sampling-Verification Danger Law in Continuous Code World Models

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Abstract

In the Code World Model (CWM) paradigm, a large language model synthesizes an executable world model that a classical planner searches, and the model is accepted when it matches sampled transitions. In discrete games, a companion paper showed this gate certifies the wrong thing: a model can pass at 100% transition accuracy and still lose systematically at play, following a quantitative law, danger = play_cost × $(1 - \text{rarity})^N$ (Aguilar Martín, 2026). This paper moves the question to continuous state spaces, where the model-based RL literature holds that world-model error is *pervasive* rather than localized. The law transfers: its ingredients are measure-theoretic, and we prove that a smooth (Lipschitz) truth/model pair cannot realize the localized verified-but-wrong geometry at all (a geometric statement; Section 3) — the premise is realizable exactly where continuous systems are hybrid, so an omitted mode is a rare rule. On two minimal hybrid instruments (a cart with an inelastic wall; a pendulum with a hard stop), the gate misses the omitted mode at the predicted $(1 - r)^N$ rate and the mode-blind planner is exploited to below-random return at every knob; a planner that merely checks its own predictions against observations collapses this exploitation to a bounded first-contact transient. With real LLM synthesis, every sample that missed the mode yielded a fully certified, fully blind, fully exploited model; when the sample contained the mode, GPT-5.x repaired the exact rule — unlike the discrete setting — while a cross-family spot-check repaired none (0/2 per instrument): repair-from-data is model-dependent, while identifiability is a property of the sample. Localization — both the danger and the repair — is a representational property of *code*.

1 Introduction

The companion paper (Aguilar Martín, 2026) established, in discrete games, that transition accuracy on randomly sampled play-throughs is the wrong adequacy criterion for a world model used in planning, and quantified the failure: danger = play_cost × $(1 - \text{rarity})^N$, with the gate-miss factor exact under i.i.d. sampling. Its related-work discussion closed with a promissory note: the rare-rule gap is a localized, discrete failure that state-accuracy metrics mask by dilution — a point worth revisiting in continuous settings, where the model-based RL literature (Lambert et al., 2020; Hafner et al., 2020; Janner et al., 2019) treats world-model error as pervasive and compounding rather than localized and pivotal.

This paper is that revisit. The question is not whether continuous world models can be wrong — that literature is mature — but whether the specific *verified-but-wrong* geometry of the discrete result exists in continuous state spaces: a model that passes a sampling gate cleanly, is (in a precise sense) almost-everywhere correct, and is still exploited catastrophically by the planner that trusts it.

The answer is yes, and the continuous version of the story is in one respect cleaner than the discrete one. The discrete headline had two components: (a) a provable one (when the rare rule is absent from the sample, no learner can infer it — an identifiability event with probability exactly $(1-r)^N$) and (b) an empirical one (the LLMs tested did not infer the rule *even when it was present* — "translation, not inference"). In the continuous instrument the second component **vanishes** for the models where we measured it: GPT-5.x infers an omitted hybrid mode from a handful of boundary-crossing transitions, writing the exact global rule. What remains provable is exactly the gate-miss and identifiability propositions (Section 3): with the mode absent from the sample — the $(1-r)^N$ event — no learner can recover it from the sample alone (a prior or the specification could still supply it; Proposition 2). That the loop never accepted a wrong mode-present artifact is an empirical, code-inspected regularity for the models tested (GPT-5.x repairs every revealed mode; Qwen stalls and the gate refuses its patches), not a theorem. The danger law here is not just exact; empirically, for the models tested, it is exhaustive.

Contributions:

1. **Theory transfer, and one new piece.** The gate-miss proposition, the identifiability proposition, and the play-cost upper bound via query-hit mass transfer to continuous state spaces with only notational change — they are measure-theoretic (Section 3) — and a corollary ties the bound to the paper's play_cost normalization (Corollary 1). The genuinely discrete ingredient of the companion paper is the *localization premise*, and we prove it requires unbounded local Lipschitz structure: for L-Lipschitz truth and model differing by η at a state-action point, the disagreement region contains a metric ball of radius $(\eta-\varepsilon)/2L$ in $S \times A$ (Proposition 4). Hybrid mode boundaries are exactly where this obstruction disappears — an omitted mode is a rare rule.

2. **Two minimal hybrid instruments** — cart-with-wall (linear off-mode plant) and pendulum-with-stop (nonlinear) — on which the full discrete phenomenology is measured with the same harness and no per-instrument re-calibration: the threshold law with the whole $(1-r)^N$ elbow inside the knob sweep, knob-invariant play_cost ≈ 1 (the exploited planner scores below random, pinned against the phantom mode all episode), and the reach mechanism in its cleanest form (Section 4). Three instrument-design lessons of independent value are recorded (Section 2.3).

3. **A pinned-integrator contract and a measured axis separation.** The pinned-integrator synthesis contract makes correct code float-exact — a property of code exactness independent of the gate's tolerance — so the synthesis gate (Section 7) can run at $\varepsilon = 10^{-9}$. Separately, at a deployment-realistic $\varepsilon = 0.01$ (Section 5), the gate rejects supra-tolerance global bias on every rollout, passes sub-tolerance bias harmlessly, and misses the hard mode at a rate that closely tracks the predicted $(1-r)^N$ (within sampling noise at gate scale; Section 5). A smooth localized perturbation of comparable rarity has ≈ 0, at higher amplitude *negative*, play cost.

4. **A measured planner-side mitigation** (Section 6): distrust-region replanning — the planner checks its model's predictions against observed transitions and fences the refuted predictions — collapses the pinned-forever, below-random exploitation to a bounded first-contact transient on all 11 knob rows across both hand-written instruments, at zero cost when the model is right (bit-identical to plain MPC, tested). The fence exploits the hard-boundary structure of these modes (Section 6); the gate-side law is untouched: the gate still certifies a wrong model.

5. **The synthesis result: danger collapses to pure identifiability** (Section 7). Real-LLM arms (GPT-5.x, 20 seeds/cell) with the identifiability event logged per seed: mode absent

$\Rightarrow$ 20/20 verified-blind-exploited (regret 0.999, Wilson lower bound 0.84); mode present $\Rightarrow$ repaired to the exact global rule (both sizes 10/10, large in ≤ 1 iteration), never a wrong artifact accepted. A Qwen cross-family spot-check reproduces the blind-exploited event but repairs neither wall-present seed — repair-from-data is model-dependent, while identifiability is a property of the sample, not the model. The whole result replicates on a second, nonlinear instrument (a pendulum with an angular hard stop): GPT-5.x repairs 62/62 mode-present seeds and every mode-absent seed is blind and exploited.

6. **Smooth learners cannot localize** (Section 8): a closed-form linear fit on mode-free data passes both gates fully blind (identifiability is learner-independent); on mode-containing data it is tilted twelve orders of magnitude off-mode by four contact rows in 3200 while still missing the mode. Both the danger geometry and the repair capability are representational properties of code.

Scope up front: two 1D instruments, one planner family (random-shooting MPC), 20 seeds per headline synthesis cell across the two instruments (both GPT-5.x sizes; plus the caught cell on the pendulum), a 3-seed cross-family spot-check per instrument (not full sweeps), and a probe-grade MLP. Section 10 gives the honest assessment and the full seed accounting.

2 The instruments

This section specifies the hand-written hybrid instruments and the planner that every later experiment reuses, and records the design lessons learned while calibrating them.

2.1 Cart-with-wall

State (x, v); action $a \in [-a_{\max}, a_{\max}]$; semi-implicit Euler with fixed dt:

$$a \leftarrow \mathrm{clamp}(a), \qquad v' = v + (\mathrm{gain} \cdot a - \mathrm{drag} \cdot v)\, dt, \qquad x' = x + v'\, dt,$$

with defaults $dt = 0.1$, gain $= 3$, drag $= 0.3$, $a_{\max} = 1$. The **hybrid mode** is an inelastic wall at x_{wall}: if $x' \geq x_{\mathrm{wall}}$, the next state is exactly $(x_{\mathrm{wall}}, 0)$. The **blind model** is the same code path with the wall branch removed — the hand-written on-manifold proxy for a CWM synthesized from a spec that omits the wall, *bit-exact off-mode by construction* (tested: exact float equality on wall-free trajectories). We call such a model **mode-blind** (the canonical term; instrument-specific synonyms like "wall-blind", or "blind" for short, mean the same thing). Reward is two sigmoid plateaus: a small reachable one on the left (0.3, at $x \leq -6$) and a large one on the right (1.0, at $x \geq 12$) whose approach every swept wall position blocks. Episodes are 80 steps from $x_0 \sim U(-0.5, 0.5)$, $v_0 = 0$.

The design requirements mirror the companion paper's material-at-cap instrument (paper 1's discrete instrument: a rare resource-cap rule likewise omittable from sampled play): (a) random rollouts rarely fire the mode — the wall position is the **rarity knob** (r sweeps $0.331 \to 0.0020$ over $x_{\mathrm{wall}} \in [2, 10]$; Table 1); (b) the omission is *exploited*, not merely mispredicted — the wall-blind model predicts coasting through the wall toward the large plateau, so the planner it advises drives right and is pinned; (c) the truth planner's optimal play differs qualitatively — it goes left.

The second instrument (pendulum with a hard angular stop; nonlinear gravity term) is described with its results in Section 4.1.

2.2 Planner and play_cost

The planner is model-predictive control (MPC) by random shooting (Nagabandi et al., 2018; Chua et al., 2018): at each step, sample candidate action sequences (piecewise-constant blocks plus the three constant sequences $\{-a_{\max}, 0, +a_{\max}\}$), roll each out on the *model*, take the best first action, replan. The planner is a deterministic function of its model's responses and the seed, so the play-cost upper bound via query-hit mass applies verbatim (Section 3; Corollary 1 ties that bound to the normalization below). Single-agent control makes play_cost a normalized regret, cleaner than a two-player arena (no opponent confound):

$$\text{play\_cost} \;=\; \frac{J_{\text{truth}} - J_{\text{model}}}{J_{\text{truth}} - J_{\text{rand}}},$$

where J_{truth}, J_{model}, and J_{rand} are the returns of the truth-planner, the model-planner (written J_{blind} in the tables when the model is the blind one), and the uniform-random policy, all measured in the true environment on paired seeds. The blind planner can score *below random* (it is actively exploited), so the normalized value can exceed 1; we report it unclamped.

2.3 Three instrument-design lessons

Recorded because they will bite any continuous CWM study:

1. **I.i.d. per-step candidate sampling silently removes the mode from imagination.** With i.i.d. candidates, imagined displacement is diffusive; no sampled sequence reaches distant reward within the horizon, so the truth model and the blind model rank all candidates *identically* and the wall never enters imagination — the two arms become indistinguishable not because the models agree but because the planner never queries where they differ. Piecewise-constant blocks plus constant candidates fix it. The planner's *query* distribution, not just its trajectory distribution, is part of the instrument.

2. **Point (Gaussian) reward lodes demand braking finesse that random shooting lacks**; sigmoid plateaus remove the parking problem and give clean imagined-value margins in both directions.

3. **The plant's drag time-constant must sit well inside the planning horizon**, or no arm can act on the reward at all (our first calibration had $\tau = 1/\text{drag} = 10\,\text{s}$ against a $3\,\text{s}$ horizon; nothing moved).

3 Theory: what transfers, what changes, what is new

Notation: a verification gate draws N i.i.d. rollouts from the gate policy ρ (uniform-random actions from the initial-state distribution) and accepts a model $\hat{f}$ if it matches the truth f within ε in sup-norm on every visited transition.

Measure-space setup. Fix a horizon T. A trajectory is a point of $(S \times A)^T$, equipped with the product Borel σ-algebra. The gate policy ρ together with the transition kernel induced by f defines a trajectory law P_ρ on this space (the initial-state distribution pushed forward through ρ and f). The query measure $\mu_{\text{query}}(E)$ (the *query-hit mass*) is the probability, under the planner's trajectory law, that at least one model query during the episode lands in E. We assume throughout that the critical region R, the disagreement region E, and the reward and dynamics maps are Borel measurable, so every probability below is well defined. With this in place the discrete arguments of Aguilar Martín (2026) transfer with only notational change.

Proposition 1 (gate miss; transfers verbatim)**.** *Let R be any measurable set of rollouts ("the critical event"; here: the rollout fires the wall mode) with $r = P_\rho(R)$. The probability that N i.i.d. gate rollouts all avoid R is exactly $(1-r)^N$.*

Nothing in the companion paper's proof uses discreteness; the event is Bernoulli(r) and the draws are i.i.d.

Proposition 2 (identifiability; transfers verbatim)**.** *Condition on the miss event of Proposition 1. Let $M \subseteq S \times A$ be the mode region (for the cart the clamp fires depending on (s, a) through $x' \geq x_{\text{wall}}$), and let $\hat{f}_1, \hat{f}_2$ be any two models that agree on $(S \times A) \setminus M$. On the miss event the sample visits no transition in M, so $\hat{f}_1$ and $\hat{f}_2$ produce identical outputs on every sampled input; hence every sample-measurable score (the gate, a likelihood, a refinement objective) is constant across the two, and no such score can distinguish them — preference for the correct one must come from the prior or the specification. This holds for* any *learner — LLM, linear regression, MLP — a point Section 8 instantiates empirically.*

Proposition 3 (play-cost upper bound; transfers verbatim)**.** *Assume returns are normalized to* $[0, 1]$ *(WLOG, by rescaling $J \mapsto (J - J_{\min})/(J_{\max} - J_{\min})$; equivalently carry an explicit factor $J_{\max} - J_{\min}$ throughout). For any planner that is a deterministic function of model responses and a seed, $|J(f) - J(\hat{f})| \leq \mu_{\text{query}}(E)$, where E is the disagreement region and μ_{query} the probability (the query-hit mass) that the planner queries its model on E during an episode.*

Proof sketch (coupling). Couple the two runs on a common seed. On the event that no query lands in E, f and $\hat{f}$ return identical responses to every query the planner issues (they agree off E), so with shared rng the two runs select the same action sequence and traverse the same states, realizing the same return. The realized returns can therefore differ only on the complementary event, of probability $\mu_{\text{query}}(E)$, where normalized returns differ by at most 1; taking expectations gives $|J(f) - J(\hat{f})| \leq \mu_{\text{query}}(E)$. MPC's imagined rollouts are the queries. (The argument is the coupling of Aguilar Martín (2026), recorded here rather than deferred.)

The raw J reported in the tables (e.g. $J_{\text{truth}} = 17.77$) is this normalized quantity rescaled by $J_{\max} - J_{\min}$, so the bound is a statement about the normalized return. The paper's reported play_cost, however, divides by $J_{\text{truth}} - J_{\text{rand}}$ (Section 2.2), *not* by $J_{\max} - J_{\min}$; the following corollary states the relationship rather than silently identifying the two.

Corollary 1 (play_cost saturation)**.** *With* $|\text{play\_cost}| = |J(f) - J(\hat{f})|/(J_{\text{truth}} - J_{\text{rand}})$*, play_cost as defined (signed) in Section 2.2, Proposition 3 in raw units ($|J(f) - J(\hat{f})| \leq \mu_{\text{query}}(E)\,(J_{\max} - J_{\min})$) gives*

$$\text{play\_cost} \;\leq\; \mu_{\text{query}}(E)\,\frac{J_{\max} - J_{\min}}{J_{\text{truth}} - J_{\text{rand}}},$$

so the bound reads $\text{play\_cost} \lesssim \mu_{\text{query}}(E)$ *exactly when $J_{\text{rand}} \approx J_{\min}$ and $J_{\text{truth}} \approx J_{\max}$ — the random policy near the reward floor, the truth planner near the ceiling. Both instruments are in this regime: on the cart $J_{\text{rand}} = 0.53$ against a pinned $J_{\text{blind}} \approx 0$ and $J_{\text{truth}} = 17.77$; on the pendulum $J_{\text{rand}} = 0.06$ and $J_{\text{truth}} = 20.08$. The blind planner queries the wall region in every episode, so $\mu_{\text{query}}(E) \approx 1$ and the normalized bound is saturated —* $\text{play\_cost} \approx 1$*, as observed in Sections 4–5.*

What does not transfer. The companion paper's coverage certificates enumerate finite information-set spaces. Continuous state spaces admit no such enumeration; covering-number analogues under Lipschitz assumptions are left open (Section 10).

What is new: the localization premise is a theorem-shaped obstruction.

Proposition 4 (smoothness forbids localized error)**.** *Let $f, \hat{f} : S \times A \subseteq \mathbb{R}^d \times \mathbb{R}^m \to \mathbb{R}^d$ be L-Lipschitz on the joint state-action space in the sup-metric on $S \times A$, with $L \in (0, \infty)$ (i.e. $L = \max(\mathrm{Lip}\, f, \mathrm{Lip}\, \hat{f})$). Suppose $\|f(s_0, a_0) - \hat{f}(s_0, a_0)\|_\infty = \eta$ at some $(s_0, a_0) \in S \times A$. Then for any tolerance $\varepsilon < \eta$, the disagreement region $E_\varepsilon = \{(s, a) \in S \times A : \|f(s, a) - \hat{f}(s, a)\|_\infty > \varepsilon\}$ contains the open metric ball $B\big((s_0, a_0), \frac{\eta-\varepsilon}{2L}\big) \cap (S \times A)$ (the intersection matters only when (s_0, a_0) lies near $\partial(S \times A)$). If instead $L = 0$ then $f - \hat{f}$ is constant and E_ε is all of $S \times A$.*

Proof. $g = f - \hat{f}$ is $2L$-Lipschitz on $S \times A$, so for $\|(s, a) - (s_0, a_0)\|_\infty < (\eta - \varepsilon)/2L$ we have $\|g(s, a)\|_\infty \geq \eta - 2L\, \|(s, a) - (s_0, a_0)\|_\infty > \varepsilon$; the inequality is strict, so the ball is open. The $L = 0$ case is immediate: $g \equiv g(s_0, a_0)$ with $\|g(s_0, a_0)\|_\infty = \eta > \varepsilon$. □

The disagreement region E_ε lives in the same joint state-action space $S \times A$ as the query region E of Proposition 3, so the ball is *directly* a lower bound on the query-relevant region the planner can hit — no fixed-action slice argument is needed to bring the two together. (Restricting to a fixed action a_0 recovers the state-slice form: the open ball of radius $(\eta - \varepsilon)/2L$ in $S \times \{a_0\}$ lies in E_ε, the version convenient when reasoning about a single instrument's mode.)

Read as its contrapositive: a model error that is *large somewhere* (large enough to matter at play) and *invisible at tolerance ε outside a metrically tiny region* requires L large — in the limit, a discontinuity. This is exactly the sense in which the discrete localization premise ("the wrong model is exact off the rule region") is discrete: it is unsatisfiable by smooth truth/model pairs, up to the ball-radius quantification. Two consequences structure the paper: (i) the natural continuous home of the danger geometry is **hybrid dynamics** — contacts, hard stops, saturations, regime switches — where the truth itself has unbounded local Lipschitz constant across a mode boundary (our wall: v jumps to 0), and an omitted mode is precisely a rare rule; (ii) the premise is *representationally* available to programs (an omitted or added branch is bit-exact off the omitted case) and not to smooth learned models, a distinction Section 8 measures.

Remark 1 (metric, not measure). The ball in Proposition 4 is metric, and its *probability* under the gate's visitation measure can still be small — smoothness bounds how spatially concentrated an error can be, not how often the gate visits it. The proposition removes the *exact* localization premise for smooth pairs; the $(1 - r)^N$ mechanism itself is representation-independent and applies to any critical region of small visitation measure.

4 The mechanism and the threshold law

This section measures the danger law's full phenomenology on the hand-written instruments, before any LLM enters the loop. All numbers CPU-only, with the hand-written blind model as the on-manifold proxy (`scripts/continuous_reach.py`; 3000 rarity rollouts and 20 MPC episodes/arm per knob; Wilson 95% CIs (Wilson, 1927) in the versioned results).

Readings: **the threshold law again** — danger ≈ 0 while the wall sits inside the random-rollout envelope, rises through the elbow, and plateaus at full play_cost, with the entire elbow inside the sweep (Figure 1). **play_cost is knob-invariant and** ≈ 1 (exceeding 1 at every knob except the largest, where far-plateau leakage gives 0.977): the blind planner is not merely uninformed but

x_{wall}	rarity	J_{truth}	J_{blind}	J_{rand}	play_cost	blind hit	truth hit	d@20	d@40	d@80
2	0.3313	17.77	0.00	0.53	1.031	1.00	0.00	0.000	0.000	0.000
3	0.2193	17.77	0.00	0.53	1.031	1.00	0.00	0.007	0.000	0.000
4	0.1430	17.77	0.00	0.53	1.031	1.00	0.00	0.047	0.002	0.000
5	0.0843	17.77	0.00	0.53	1.031	1.00	0.00	0.177	0.030	0.001
6	0.0503	17.77	0.00	0.53	1.031	1.00	0.00	0.367	0.131	0.017
8	0.0127	17.77	0.02	0.53	1.030	1.00	0.00	0.798	0.619	0.372
10	0.0020	17.77	0.94	0.53	0.977	1.00	0.00	0.938	0.901	0.832

Table 1: The danger curve on the wall-position knob. "blind/truth hit" = fraction of episodes in which that planner's trajectory fires the wall mode; d@N = play_cost $\cdot (1-r)^N$.

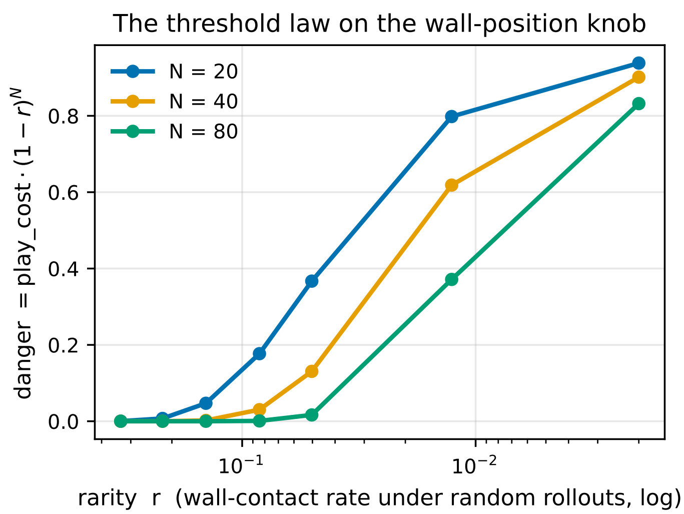

Figure 1: The threshold law on the wall-position knob (Table 1 rendered): danger ≈ 0 while the wall is inside the random envelope, rises through the elbow, plateaus at full play_cost; N shifts the threshold.

exploited — MPC on the wall-less model plans into the phantom region, is pinned at the wall (final $x = x_{\text{wall}}$ exactly, contact rate 1.00 at every knob), and replans the same doomed plan every step, for the entire episode, scoring below random (0.00 vs 0.53). The 0.977 at $x_{\text{wall}} = 10$ is the sigmoid tail of the far plateau leaking $J_{\text{blind}} = 0.94$; the mechanism is unchanged. **The reach mechanism** appears in its cleanest form (Figure 2), with the query/trajectory distinction now visible. Gate-miss exactness is re-verified in-tests and again at gate scale in Section 5.

4.1 Robustness: the same law on a nonlinear plant

The cart's off-mode dynamics are linear, which is convenient for Section 8 but invites the worry that the phenomenology depends on it. A second instrument — a pendulum (gravity term $\sin\theta$, $\theta = 0$ hanging down) with a hard angular stop, same interface, same MPC, same two-plateau reward on θ — reproduces the identical picture with *no* re-calibration (rarity is natural here: gravity confines the random walk near the bottom, so climbing to the stop is rare):

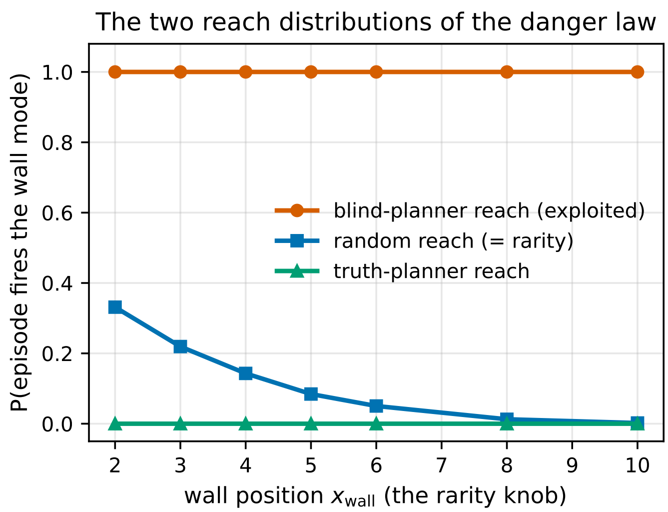

Figure 2: The two reach distributions of the danger law: the exploited planner's mode reach is flat at 1.00 across the knob; random reach (= rarity) falls two orders of magnitude; the truth planner's trajectory reach is 0.00 — the mode lives on its *query* distribution (Proposition 3), not its path.

θ_{stop}	rarity	J_{truth}	J_{blind}	play_cost	blind hit	d@40
0.8	0.2970	20.08	0.01	1.002	1.00	0.000
1.0	0.1277	20.08	0.03	1.002	1.00	0.004
1.2	0.0527	20.08	0.05	1.000	1.00	0.115
1.4	0.0193	20.08	0.12	0.997	1.00	0.457
1.6	0.0073	20.08	0.26	0.990	1.00	0.737
2.0	0.0000	20.08	1.23	0.942	1.00	0.942

Table 2: Pendulum-with-stop: threshold law, knob-invariant exploitation (pinned at the stop in every episode at every knob), truth planner untouched by the mode. The mechanism does not care that the plant is nonlinear — only that the mode is hard and rare under the gate's measure.

5 Axis separation: the gate fails only where the law says it can

The classic continuous-model failure axis is pervasive sub-tolerance error; the danger law's axis is a localized hard mode. A tolerance gate ($\varepsilon = 0.01$, deployment-realistic) must be shown to fail *only* on the second axis, and only at the $(1-r)^N$ rate. Five arms, one table (`scripts/continuous_axes.py`; reveal-rarity = P(a random rollout contains a transition where truth and model differ $> \varepsilon$), 2000 rollouts; pass@40 over 300 independent N=40 gates; 20 MPC episodes/arm). For the hard mode, the mode-firing rarity of Tables 1–2 and the reveal-rarity here coincide as events — the mode error exceeds every deployment-realistic ε (the ε-sweep below), so a rollout fires the mode iff it reveals a disagreement — which licenses the shared symbol r.[1]

Readings (Table 3, rendered as the danger quadrant in Figure 3): **gate exactness at gate scale** — measured pass@40 matches $(1-r)^{40}$ in both wall rows (0.003 vs 0.0026; 0.667 vs 0.605, the prediction marginally below the 300-gate Wilson lower bound of 0.6115 — sampling noise at that gate count; see the ε-sweep note below); the proposition is the observed acceptance rate,

[1] The small numeric difference for wall@4 — mode-firing rarity 0.1430 (3000 rollouts, Table 1) vs. reveal-rarity 0.1385 (2000 rollouts, Table 3) — is resampling noise on the same event, not a different event.

arm	reveal-rarity	$(1-r)^{40}$	pass@40 measured	play_cost	d@40
wall@4 omitted	0.1385	0.0026	0.003	1.031	0.0027
wall@8 omitted	0.0125	0.6046	0.667	1.030	0.6227
drag bias $\times 1.03$ (sub-ε)	0.0000	1.0000	0.997	0.000	0.0000
drag bias $\times 2.0$ (supra-ε)	1.0000	0.0000	0.000	0.000	0.0000
C^∞ bump@4, amp 0.5	0.1875	0.0002	0.000	0.000	0.0000
C^∞ bump@4, amp 1.0	0.2085	0.0001	0.000	-0.745	-0.0001

Table 3: Axis separation at $\varepsilon = 0.01$, $N = 40$; d@N = play_cost $\cdot (1-r)^N$, as in Table 1. Danger lives in one quadrant only: rare $\wedge$ hard-mode.

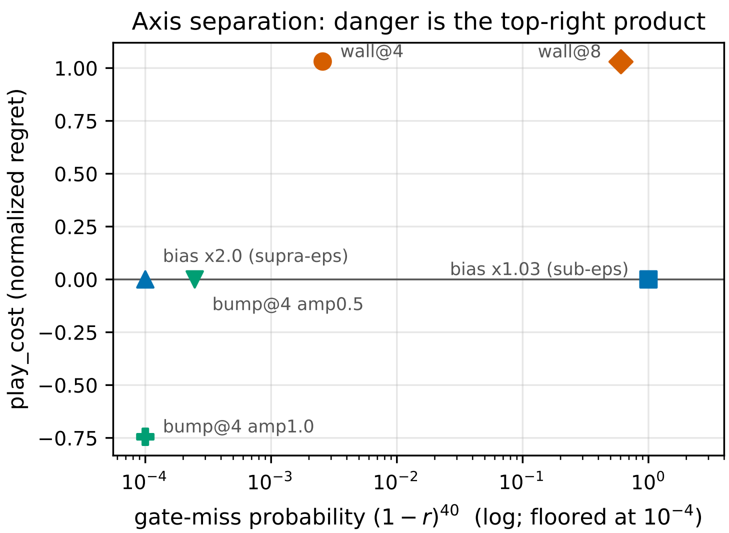

Figure 3: Table 3 as the danger quadrant: gate-miss probability (log) vs play_cost. Only the rare hard mode reaches the top right; danger is the product of the coordinates.

not asymptotic decoration. **The gate polices the pervasive axis**: a global drag bias above tolerance is revealed on every rollout and never accepted; a sub-tolerance bias is accepted (0.997; the 0.3% is the extreme velocity tail) and is harmless at play. Verified-and-fine is a real cell, and the gate finds it. **Smoothness kills consequence, not detectability**: the C^∞ drag bump at the wall's location has comparable rarity to the wall (0.19 vs 0.14 — it is just as *detectable*, confirming Proposition 4 is about error geometry, not about hiding from the gate) yet play_cost 0.000 at amplitude 0.5. At amplitude 1.0 play_cost turns *negative* (-0.745): the truth planner, seeing the slowdown near its horizon edge, is over-pessimistic and often settles for the small plateau, while the bump-blind planner pushes through and wins. A smooth localized omission produces planner-side timing effects of ambiguous sign; only the hard mode produces the one-way exploitation geometry (pinned, forever, below random).

Is $\varepsilon = 0.01$ a special setting, or is the axis separation itself ε-invariant? A sweep over $\varepsilon \in \{10^{-9}, 10^{-6}, 10^{-4}, 10^{-3}, 10^{-2}, 3{\times}10^{-2}, 0.1, 0.3\}$ (`scripts/continuous_eps_sweep.py`; full table in `docs/EXPERIMENTS.md`) answers: ε-invariant. Mode-arm reveal-rarity is flat across the entire deployment-realistic range — on the cart, `wall@8` is bit-identically flat through the whole grid including $\varepsilon = 0.3$, and `wall@4` only dips slightly (never widens) at the top of the grid. The pervasive bias arms switch sharply at their own error scale on both instruments instead.

ε	wall@8 rarity	bias ×1.03 rarity	bias ×2.0 rarity
10^{-6}	0.0125	1.0000	1.0000
10^{-2}	0.0125	0.0000	1.0000
0.1	0.0125	0.0000	0.0040
0.3	0.0125	0.0000	0.0040

Table 4: Reveal-rarity vs. ε (cart). The mode arm is flat across the whole grid; the pervasive arms switch at their own error scale.

The pendulum replicates both halves: its mode arms are flat too (only a slight dip at the top of the grid — `stop@1.0` rarity 0.1410 at $\varepsilon \leq 3{\times}10^{-2}$, dipping to 0.1400 at $\varepsilon = 0.1$ and 0.1240 at $\varepsilon = 0.3$), and its bias arms switch at the same error-scale boundaries as the cart's. pass@40 $\approx (1-r)^{40}$ continues to hold for the mode arms at every ε in the grid, on both instruments (honesty note: at `wall@8` the closed-form 0.6046 sits marginally below the empirical 300-gate Wilson 95% CI lower bound of 0.6115 — sampling noise at that gate count, reported plainly rather than as exact agreement). `play_cost` is not re-measured across ε: the model under test does not depend on the gate's tolerance, so play behavior is ε-independent by construction — the sweep varies only what the gate can see. The gate's ε is a pervasive-error dial, not a mode-detection dial: tightening it cannot catch the hard mode, and loosening it does not widen the hole.

6 Mitigation: the exploitation is planner-mediated

The exploitation measured in Sections 4 and 5 is planner-mediated, not model-mediated, and a planner-side fix collapses it without touching the model or the gate — this does **not** contradict the danger law (the gate still certifies a wrong model; Proposition 2 is untouched). Like those sections, this one uses the hand-written instruments and blind models, not LLM synthesis. Distrust-region replanning (`src/cwm/continuous/mitigation.py`, strictly additive) compares the model's prediction against the observed transition after every real step (tol $= 10^{-6}$); a disagreement records the *position of the model's refuted prediction* — not the pre-state — as a one-sided fence, since false predictions always lie on or beyond the mode boundary. While scoring a candidate rollout, the first imagined step whose position interval crosses a fence's ε-band ($\varepsilon = 0.25$ cart, $\varepsilon = 0.1$ pendulum) truncates the rollout — reward kept, everything downstream dropped — which makes the fence leap-proof at any imagined speed; candidates are then ranked by (truncated return, distance to the nearest fence), which structurally prefers the real side. With zero violations this is bit-identical to plain MPC by construction (tested bitwise) — mitigation costs nothing when the model is right. Only this design is undodgeable: three earlier attempts (pre-state flee-balls; full-state point fences, dodged by the planner probing crossing velocities) were each defeated by the argmax planner acting as an adversary against the fence. The collapse is scoped to **hard-boundary hybrid modes**: the one-sided fence works precisely because a refuted prediction lies on or beyond the mode boundary, so fencing its far side cannot cut off any real trajectory. This structural fact is what these wall/stop instruments supply; less structured failure modes (soft or moving boundaries, errors not confined to a hard stop) do not obviously admit such a fence and are untested — this is a hard-boundary mitigation, not yet a general planner-side one.

The collapse is large at every knob: `pc_blind` stays pinned at ≈ 0.94–1.03 everywhere (established in Section 4, Tables 1 and 2), while `pc_mit` never exceeds 0.81. Exactly one violation suffices to fence the mode on *every* one of the 11 rows — the mitigated planner must touch the mode once, which is identifiability operationalized: you cannot avoid what you have never seen. The residual

knob	pc_blind	pc_mit	first-contact step
cart (x_{wall})			
2	1.031	0.290	11.6
4	1.031	0.446	16.9
6	1.031	0.578	21.3
8	1.030	0.699	25.1
10	0.977	0.806	28.7
pendulum (θ_{stop})			
0.8	1.002	0.113	7.0
1.0	1.002	0.129	8.1
1.2	1.000	0.143	9.0
1.4	0.997	0.160	10.0
1.6	0.990	0.177	11.0
2.0	0.942	0.212	13.0

Table 5: Mitigation sweep (20 episodes/knob, both instruments). pc_blind / pc_mit = play_cost of the blind planner under plain MPC / under distrust-region mitigation. Full 11-row table with contact rates and violation counts in `docs/EXPERIMENTS.md`.

`pc_mit` is the honest cost of that unavoidable first contact, and it grows with the lure distance, read off the first-contact step: cart 0.290 → 0.806 as first contact goes 11.6 → 28.7 of 80 steps (knob 2 → 10); pendulum 0.113 → 0.212 as first contact goes 7.0 → 13.0 (knob 0.8 → 2.0). The cart knob = 10 row is the least favorable in the sweep (`pc_mit` 0.806, close to blind's 0.977) because the transient consumes most of the horizon — but the blind planner stays pinned *forever* there (J 0.94 of J_{truth} 17.77) while the mitigated planner escapes and recovers most of the horizon (J 3.88): the normalized cost looks modest, the actual return does not.

7 LLM synthesis: the danger collapses to pure identifiability

Real-LLM arms (`scripts/continuous_danger_synthesis.py`; Azure GPT-5.x mini and large; $N = 40$ training rollouts which double as the gate, as in the companion paper's sweep; $\varepsilon = 10^{-9}$ pinned-integrator gate; **20 seeds/cell on the headline $x_{\text{wall}} = 8$ cell, both sizes** (the companion paper's standard); 6 MPC play episodes/seed; per-seed JSON with the synthesized code versioned in the repository). The contract pins the integrator (Section 2.1's equations, stated in the spec text, constants generated from the environment instance so they cannot drift); the *full* arm includes the wall clause, the *incomplete* arm omits it. Crucially, each seed logs whether the wall fired in its training sample — the identifiability event that the companion paper could not condition on post hoc.

Full arm (both sizes, 40 seeds): gate 1.000 in 0 refinement iterations, wall probes exact, play at truth parity — every seed. The pinned-integrator premise holds with a real LLM: correct synthesis is float-exact through the sandbox, so $\varepsilon = 10^{-9}$ costs nothing and the tolerance axis is fully disarmed. As in the discrete setting, given the rule, the model translates it perfectly.

Incomplete arm. Three-way structure:

1. **Wall absent from the sample** (the $(1-r)^N$ event; 10/20 seeds at $x_{\text{wall}} = 8$ for each size, consistent with $(1-0.0125)^{40} \approx 0.60$): every such seed — **20/20 across mini and large** —

passed the gate at 1.000, fully wall-blind on the probes, and was exploited at play: pinned at the wall, contact rate 1.0, **play_cost 0.999** (Wilson 95% lower bound 0.84 combined). This is the discrete headline, synthesized end-to-end in a continuous CWM: a verified, almost-everywhere-exact model that performs worse than random.

2. **Wall present, repaired:** the LLM does *not* stay blind. It reads the failing transitions and writes the true global rule — `if x2 >= 8.0: return [8.0, 0.0]` (or the equivalent `if x2 > 8.0: x2 = 8.0`) — not a curve fit. At 20 seeds both sizes repaired **every** wall-present seed: **large 10/10 in 0–1 iterations** (two from the synthesis examples alone), **mini 10/10 in 0–5 iterations**. This is the divergence from the discrete setting, where rules demonstrated by example transitions were persistently not learned (translation-not-inference). A numerically-manifested discontinuity is learnable from data in a way a symbolic game rule was not.

3. **Wall present, not repaired:** at 20 seeds on the headline cell GPT-5.x produced **no stalls**, but stalls are real where they occur (the 5-seed $x_{\text{wall}} = 4$ cell; the Qwen cross-family arm below). They are not near-misses of the rule but **superstitious local patches**: e.g. `if abs(x2 - 8.0) <= 0.15 and abs(v2) <= 1.1: x2 = 8.0`, or Qwen's `if x2 >= 8.0 and v2 <= 0.0: ...` — clamps fitted to the *observed manifestation* of the mode (low-speed, near-wall contacts), which mispredict other approaches to the wall. **The gate rejected every one of them** (gate 0.49–0.999, never 1.000).

Cross-family spot-check (HF router, `Qwen/Qwen3-Coder-30B-A3B-Instruct`, 3 seeds). The full arm is clean (3/3 gate 1.000, blind 0.0); the pinned-integrator premise is not GPT-specific. On the incomplete arm the identifiability branch reproduces (the 1/3 wall-absent seed is gate-1.000 wall-blind and exploited, play_cost 0.999), but Qwen *repaired neither* of its two wall-present seeds (gate 0.999 and 0.491, both superstitious patches the gate refused). Repair-from-data is thus *model-dependent* while identifiability is not: the wall-absent blind-and-exploited event fired for every model tried, whereas how reliably the loop recovers a *revealed* mode — and thus how much of the discrete (b)-residual it erases — varies with the synthesizer. GPT-5.x erases it entirely on this instrument; Qwen does not.

The branches compose into the paper's central claim. With the mode in the data, the synthesize–refine–gate loop behaved soundly: it either recovers the exact mode or refuses the artifact (true for every model tested — the gate never accepted a wrong wall-present artifact — an empirical, code-inspected regularity, not a theorem). With the mode absent, no loop can help (Proposition 2: the sample carries no evidence for the mode — though a prior or the specification still could supply it), and the acceptance of a blind artifact is not a failure of the LLM, the refinement, or the gate implementation — it is the sampling event whose probability is exactly $(1-r)^N$. The discrete danger law had a provable core plus an empirical residual (the rule not learned even when shown); in the continuous instrument the residual *can* vanish — it does for GPT-5.x, which repairs every revealed mode — so **the law becomes the entire failure surface** for a capable-enough synthesizer, and shrinks toward it even for a weaker one. The actionable consequence sharpens correspondingly: in this regime, spec completeness and gate-sample coverage are not two independent worries — coverage *is* the dominant worry, because the synthesis loop repairs what the sample reveals.

(Scope: 20 seeds/cell on the headline cell, both GPT-5.x sizes; the wall-absent conditional is 20/20 across sizes and consistent across three independent runs — the 5-seed first run, the 20-seed tightened run, and the 3-seed Qwen run; the wall-present repair rate is 20/20 for GPT-5.x on this cell and 0/2 for Qwen — model-dependent, small-n on the cross-family arm. LLM synthesis is stochastic across calls; the three-way structure and the identifiability conditional are stable run-to-run, per-seed iteration counts are not.)

Second-instrument robustness (pendulum-with-stop, Section 4.1, 20 seeds/cell, both sizes). The synthesis arm is not cart-only. Running the identical pipeline on the nonlinear pendulum — headline $\theta_{\text{stop}} = 1.4$ (rarity 0.019) and caught $\theta_{\text{stop}} = 1.0$ (rarity 0.128; "caught" labels a cell whose mode is common enough that the gate sample essentially always contains it, so the identifiability event essentially never fires) — reproduces every branch, including a 3-seed Qwen cross-family spot-check at the headline knob (Table 6).

cell (20 seeds each)	full	absent → blind & exploited	present → repaired (stalled)
mini $\theta_{\text{stop}} = 1.4$	20/20	9 → 9 (pc 0.995)	11 → 11 (0)
large $\theta_{\text{stop}} = 1.4$	20/20	9 → 9 (pc 0.995)	11 → 11 (0)
mini $\theta_{\text{stop}} = 1.0$	20/20	0 → —	20 → 20 (0)
large $\theta_{\text{stop}} = 1.0$	20/20	0 → —	20 → 20 (0)
Qwen $\theta_{\text{stop}} = 1.4$ (3 seeds)	3/3	1 → 1 (pc 0.995)	2 → 0 (2 stalled @0.9997)

Table 6: Pendulum synthesis cells (20 seeds/cell, both GPT-5.x sizes; 3-seed Qwen spot-check at $\theta_{\text{stop}} = 1.4$). $k \to m$ = of the k seeds in that branch, m had the stated outcome (blind & exploited, resp. repaired); the parenthetical count is stalled seeds; pc = play_cost.

Pooled across both knobs and both sizes (Table 6), every mode-absent occurrence was blind and exploited at play_cost 0.995 (Wilson 95% lower bound 0.824 pooled; per-size headline 9/9, lower bound 0.701) — the same fixed-point exploitation as the cart's play_cost ≈ 1 — and GPT-5.x repaired **62/62** mode-present seeds to the exact angular clamp (`if th2 >= theta_stop: return [theta_stop, 0.0]`), 0 stalls (Wilson 95% lower bound 0.942). Qwen reproduces the mode-absent blind-exploited event but stalls on both its mode-present seeds, the same superstitious-patch signature as on the cart. Repair is model-dependent; identifiability, being a property of the sample, is not — on this instrument too. The mechanism was already validated on two instruments (Section 4.1); the synthesis result now is as well: a nonlinear plant with an angular, not positional, hard stop reproduces the same danger law and the same repair capability, so the repair finding is not a cart artifact.

8 Smooth learners cannot localize: both halves, measured

If localization is representational, two things must be checkable on non-code learners trained on the *same data* (`scripts/continuous_smooth_probe.py`; the two most favorable smooth learners: closed-form linear least squares — off the wall, the dynamics are *exactly linear*, so this is the smooth best case — and a small tanh MLP, probe-grade):

model	trained on	off-mode err mean / max	probe err	gate 10^{-9}	gate 10^{-2}
linear-LSQ	wall-free	3.6e−15 / 1.7e−14	4.18	**PASS**	**PASS**
linear-LSQ	wall-data	1.9e−03 / 1.2e−02	4.17	fail	fail
MLP h=8	wall-free	3.5e−03 / 5.0e−02	4.20	fail	fail
MLP h=8	wall-data	6.0e−03 / 4.9e−02	4.19	fail	fail

Table 7: Smooth learners on the synthesis samples ($x_{\text{wall}} = 8$). "probe err" is the wall-region probe error (4.2 $\approx$ predicting straight through the wall).

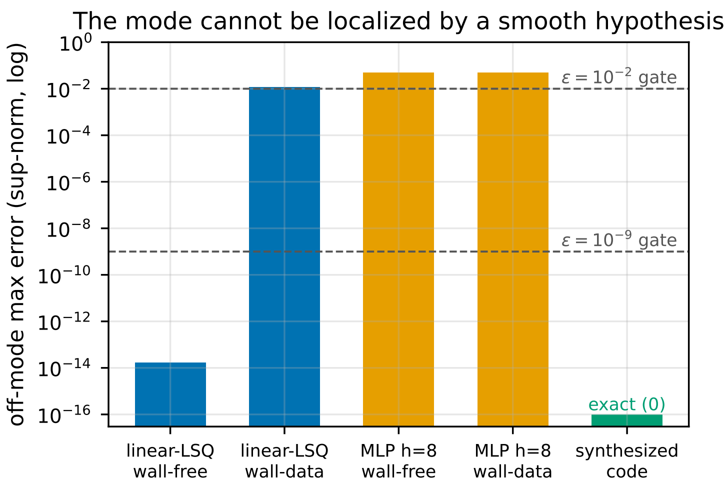

Figure 4: Off-mode max error (log) by model and training data. The wall-free linear fit sits at float noise and passes both gates (blind); four contact rows tilt it twelve orders of magnitude; synthesized code is exactly zero off-mode.

Identifiability is learner-independent, live. On the wall-free sample the linear model recovers the off-mode dynamics to 10^{-15}, **passes the** $\varepsilon = 10^{-9}$ **gate**, and is exactly as wall-blind as the synthesized blind code (probe error 4.18: it predicts straight through the wall). Proposition 2 instantiated on a second hypothesis class: in the gate-miss event, *any* accepted learner is blind. The $(1-r)^N$ hole is not an LLM property.

With the mode in the data, code and smooth part ways. Four contact rows out of 3200 tilt the linear fit by **twelve orders of magnitude** off-mode (1.7e−14 → 1.2e−02 max; Figure 4) — it fails both gates *and still* has the mode wrong (probe 4.17). The smooth hypothesis cannot put the error on the mode; it leaks everywhere (Proposition 4's geometry, observed as a least-squares tradeoff). The synthesized code, on the *same sample*, wrote the exact clamp and passed at float precision. The MLP combines both axes: a pervasive ∼5e−3 floor that no gate accepts, and a never-learned mode.

Consequence for the CWM paradigm: the paradigm *creates* in continuous control the localized failure class that the learned-model literature says does not dominate there — and, by the same representational fact, also creates the exact-repair capability that learned models lack. Code cuts both ways, and the gate's sampling coverage decides which way.

9 Related work

Objective mismatch and model exploitation in model-based RL (Lambert et al., 2020; Janner et al., 2019; Hafner et al., 2020): prediction accuracy and control performance diverge for learned continuous models; planners exploit model errors. Our contribution to that conversation is the *verified-but-wrong localized* regime, which smooth learned models cannot even represent (Proposition 4, Table 7), plus a closed-form acceptance-failure law confirmed at gate scale. *Hybrid systems:* mode detection and identification of piecewise/hybrid dynamics is classical (Paoletti et al., 2007; Bemporad and Morari, 1999); our question is not identifying the mode but what a *sampling verifier certifies* when the mode is missed, and what a planner then does. *Code World Models* (Lehrach et al., 2026) and code-as-policy (Liang et al., 2023; Gao et al., 2023) supply the paradigm; the companion paper (Aguilar Martín, 2026) is the discrete study this paper extends — transferring its provable core, showing its empirical residual (translation-not-inference) does not transfer, and adding the representational theorem separating code from smooth hypotheses. *Property-based testing and rare events* (Claessen and Hughes, 2000; Rubinstein and Kroese, 2017): the gate is random testing of a program against an oracle; $(1-r)^N$ is the standard rare-input coverage gap, here with the planner as the adversary that finds it. *Falsification and rare-event testing of hybrid and cyber-physical systems* (Annpureddy et al., 2011; Corso et al., 2021): this literature searches for inputs that drive a hybrid system to violate a specification; our gate is the passive dual — uniform random testing whose blind spot is precisely the rare mode a falsifier would hunt for, and the $(1-r)^N$ factor quantifies the coverage gap that random testing leaves. *Runtime monitoring and shielding for safe RL and control* (Alshiekh et al., 2018): a synthesized shield overrides unsafe actions using an a-priori specification; our distrust-region fence is instead deployment-time feedback, built from observed prediction failures rather than a given safety automaton, and it corrects a certified-but-wrong model rather than an unsafe policy. *Robust MPC and planning under model uncertainty* (Rawlings et al., 2017): robust and tube-based MPC guarantees performance against *bounded* model error; the failure here is orthogonal — the error is unbounded but localized to an omitted mode and is certified away by the gate, so robustness margins tuned to the off-mode error do not see it. *System identification of contact-rich and discontinuous dynamics* (Paoletti et al., 2007; Fazeli et al., 2017): estimating parameters and contact forces of rigid bodies undergoing frictional contact is a mature sysid problem; our repair-from-data result is the code analogue — given contact transitions in the sample, the synthesizer recovers the exact switching rule, whereas a smooth fit cannot localize it.

10 Limitations and Honest Assessment

Two minimal instruments, one dimension. Cart-with-wall and pendulum-with-stop are minimal by design, and Section 4.1 shows the mechanism survives a nonlinear plant; Section 7 shows synthesis does too. But both modes are single stationary boundaries in a 2-dimensional state; multi-mode, moving-boundary, and higher-dimensional instruments (2D sticky patch, contact-rich manipulation) are future work. We expect the *mechanism* to survive — it is measure-theoretic — and the *repair* finding (Section 7) has now held on both instruments, but it could still weaken as mode geometry gets harder to induce from few examples on more complex instruments.

One planner family. Random-shooting MPC with piecewise-constant candidates is the only *base* planner family studied; distrust-region replanning (Section 6) is a mitigation layered on top of it, not a second family. The mitigation claim is no longer speculative — it is measured on both instruments (Section 6, 11/11 knob rows): a planner that merely checks its own predictions against observations collapses the pinned-forever, below-random exploitation to a bounded first-contact transient, at zero cost when the model is right (bit-identical, tested), consistent with our thesis (verify on the deployment distribution) rather than against it. The claim is scoped to hard-boundary hybrid modes — the one-sided fences exploit the structural fact that refuted predictions lie on or beyond the mode boundary; less structured failure modes are untested (Section 6). The honest remaining scope is narrower still: whether other base search strategies (CEM, gradient-based shooting, tree search (Coulom, 2006; Kocsis and Szepesvári, 2006)) exhibit the same pinned-forever exploitation, and whether the same mitigation collapses it the same way, is untested.

Synthesis cells are modest. 20 seeds/cell on the headline $x_{\text{wall}} = 8$ cart cell for both GPT-5.x sizes, plus a 3-seed Qwen cross-family spot-check; the pendulum arm adds two knobs (headline and caught) at the same 20 seeds/cell, both sizes, plus its own 3-seed Qwen spot-check. The wall/mode-absent conditional is 20/20 across cart sizes and three runs (Wilson lower bound 0.84) and 18/18 pooled on the pendulum (lower bound 0.824); the GPT-5.x repair rate is 20/20 on the cart headline cell and 62/62 pooled on the pendulum. Both cross-family arms are small-n (3 seeds) and show repair is model-dependent (Qwen 0/2 mode-present on each instrument) — a single alternate family, not a sweep of models.

The MLP is a probe, not a baseline. It substantiates the representational point at h=8/pure-Python scale; a tuned modern dynamics model would have a lower floor but the same structural inability to be bit-exact off a mode at $\varepsilon = 10^{-9}$ (an argument, not yet a measurement, at scale).

Proposition 4 bounds geometry, not probability. The ball is metric; converting to gate-visitation probability needs the gate measure on that ball, which is instrument-specific. The empirical complement (Table 7's twelve-orders tilt) carries the quantitative weight here.

Coverage certificates do not transfer. The companion paper's enumeration-based guarantees have no continuous analogue in this paper; covering-number versions under Lipschitz assumptions are open.

play_cost > 1 is a normalization artifact (blind $<$ random), reported unclamped and explained; the headline claims never depend on the excess over 1.

11 Conclusion

The companion paper ended by diagnosing the gate failure as a reach-distribution shift and prescribing verification on the distribution the planner visits. This paper shows the diagnosis is not about discreteness. The gate-miss law, the identifiability argument, and the play-cost bound are measure-theoretic and survive the move to continuous state spaces intact; what the move changes is *where the localized failure can live* (hybrid mode boundaries — an omitted mode is a rare rule) and *who can express it* (programs, not smooth function classes: Proposition 4 and Table 7). On two minimal hybrid instruments the entire discrete phenomenology reproduces — threshold law, knob-invariant exploitation, the reach mechanism — with the gate's acceptance rate closely tracking $(1-r)^N$ at gate scale (within sampling noise; Section 5).

The synthesis experiment then delivers this paper's own finding: the continuous regime is the one where the danger law is *everything*. Given the mode in its sample, the LLM repairs it exactly — including from the synthesis examples alone — and when it fails to repair, the gate catches the superstitious patch; given the mode absent, every learner we tried (LLM, exact linear regression) is certified blind and the planner is exploited to below-random performance. The provable core is the gate-miss and identifiability propositions (Section 3): the mode-absent event is a hole no learner can close *from the sample* — a prior or the specification could still supply the mode — and its probability is in closed form. That the loop never accepted a wrong mode-present artifact is an empirical regularity of the models tested (GPT-5.x repairs; Qwen stalls and is refused), not a theorem. For practitioners the message is one clause sharper than the discrete one: in continuous code-world-model pipelines, *sample coverage of the mode boundaries is the whole game* — the synthesis stack will translate what it is told and repair what it is shown, and no stage of it can know about what the sample never touched.

A Reproducibility

All code is at https://github.com/JaviMaligno/code-world-models (`src/cwm/continuous/`, `scripts/continuous_*.py`); the experiment log is `docs/EXPERIMENTS.md` (sections prefixed "PAPER 2") and per-seed artifacts, including every synthesized program, are versioned under `results/`. CPU results (Tables 1–7):

```
PYTHONPATH=src python scripts/continuous_reach.py
                     # danger-curve table (tab:danger; ~2.5 min)
PYTHONPATH=src python scripts/continuous_axes.py
                     # axis-separation table (tab:axes; ~3 min)
PYTHONPATH=src python scripts/continuous_smooth_probe.py
                     # smooth-learner table (tab:smooth; ~11 s)
PYTHONPATH=src python scripts/continuous_pendulum.py
                     # pendulum table (tab:pendulum; ~2 min)
python scripts/make_paper2_figures.py     # figures from the JSONs
```

LLM arms (Azure credentials in `.env`; the runbook — commands, cost, predictions, troubleshooting — is the section "Runbook — LLM arms" of the design spec under `docs/specs/`, dated 2026-07-06):

```
PYTHONPATH=src python scripts/continuous_danger_synthesis.py mini 5
PYTHONPATH=src python scripts/continuous_danger_synthesis.py mini 5 --x-wall 4
PYTHONPATH=src python scripts/continuous_danger_synthesis.py large 5
# Pendulum synthesis arm (second instrument; Azure credentials as above)
PYTHONPATH=src python scripts/continuous_danger_synthesis.py mini 20 \
  --instrument pendulum --th-stop 1.4
PYTHONPATH=src python scripts/continuous_danger_synthesis.py large 20 \
  --instrument pendulum --th-stop 1.4
PYTHONPATH=src python scripts/continuous_danger_synthesis.py mini 20 \
  --instrument pendulum --th-stop 1.0
PYTHONPATH=src python scripts/continuous_danger_synthesis.py large 20 \
  --instrument pendulum --th-stop 1.0
# Qwen cross-family spot-checks
# (needs HF_TOKEN: HF Inference Providers router)
PYTHONPATH=src python scripts/continuous_danger_synthesis.py mini 3 \
  --compat-model "Qwen/Qwen3-Coder-30B-A3B-Instruct"
PYTHONPATH=src python scripts/continuous_danger_synthesis.py mini 3 \
  --instrument pendulum --th-stop 1.4 \
  --compat-model "Qwen/Qwen3-Coder-30B-A3B-Instruct"
# Mitigation sweep (CPU only)
PYTHONPATH=src python scripts/continuous_mitigation.py
# eps-sensitivity sweep (CPU only)
PYTHONPATH=src python scripts/continuous_eps_sweep.py
```

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